

Problem

Show that

$$\mathcal{L}^{-1}\left[\frac{4s^2 + 7s + 13}{(s+2)(s^2 + 2s + 5)}\right] = [\sqrt{2}e^{-t}\cos(2t + 45^\circ) + 3e^{-2t}]u(t)$$

Step-by-step solution

Step 1 of 1

$$F(S+2) = \frac{4S^2 + 7S + 13}{(S+2)(S^2 + 2S + 5)} = \frac{A}{S+2} + \frac{B_s + C}{S^2 + 2S + 5}$$

$$\Rightarrow 4S^2 + 7S + 13 = S^2(A+B) + S(2A+2B+C) + 5A+2C$$

Comparing coefficients:-

$$A+B=4 \Rightarrow \boxed{B=4-A}$$

$$2A+2B+C=7 \Rightarrow 8+C=7 \quad \boxed{C=-1}$$

$$5A+2C=13 \Rightarrow 5A=13+2 \Rightarrow \boxed{A=3}$$

$$B=4-A$$

$$=4-3=1 \quad \boxed{B=1}$$

$$\text{Thus :- } F(S) = \frac{3}{S+2} + \frac{S-1}{(S+1)^2 + 2^2}$$

$$= \frac{3}{S+2} + \sqrt{2} \left(\frac{S \cdot \frac{1}{\sqrt{2}} - 1 \cdot \frac{1}{\sqrt{2}}}{(S+1)^2 + 2^2} \right)$$

$$f(t) = \left\{ 3e^{-2t} + \sqrt{2}e^{-t} \cos\left(2t + \frac{\pi}{4}\right) \right\} u(t)$$

$$= \left\{ \sqrt{2}e^{-t} \cos(2t + 45^\circ) + 3e^{-2t} \right\} u(t)$$

Hence Proved.